

Session 03 (AIML) - Assignment Questions

Q1. (Basic Function)

Write a function named `welcome()` that prints the message:

"Welcome to Python Programming!"

Call the function to execute it.

Add a comment explaining why functions are useful.

Q2. (Function with Parameter)

Create a function named `greet(name)` that takes one parameter `name` and prints:

"Hello, [name]! Welcome back."

Call the function with your own name.

Also call it with a different name to show reusability.

Q3. (Default Parameter)

Write a function `show_message(message="Hello")` that prints the message.

- Call the function without passing any argument.
- Call the function again by passing a different message.

Explain the benefit of default parameters in a comment.

Q4. (Function with Multiple Parameters & Return)

Create a function `calculate_sum(a, b)` that takes two numbers and **returns** their sum.

- Call the function with two numbers and store the result in a variable.
- Print the result.

Also demonstrate the difference between `print()` inside the function and using `return`.

Q5. (Positional vs Keyword Arguments)

Write a function `student_info(name, age)` that prints the name and age.

Call this function **twice**:

1. Using positional arguments.
2. Using keyword arguments.

Add comments explaining the difference.

Q6. (Function with Return Value)

Write a function `square(num)` that returns the square of the given number.

Write another function `cube(num)` that returns the cube of the number.

In the main program, take a number as input from the user and print its square and cube using these functions.

Q7. (Recursion - Basic)

Write a recursive function `countdown(n)` that prints numbers from `n` down to 1.

Call `countdown(10)`.

Make sure to include the base case (stopping condition).

Q8. (Recursion - Factorial)

Write a recursive function `factorial(n)` that calculates and **returns** the factorial of a number.

Example: `factorial(5)` should return 120.

Test it with input from the user and print the result.

Add a comment explaining the base case.

Q9. (Scope - Local vs Global)

Write a program to demonstrate scope:

- Create a **global** variable `total = 0`
- Write a function `add_value(x)` that adds `x` to `total` (use the `global` keyword).
- Call the function multiple times and print the value of `total` after each call.

Also try to print a local variable outside its function to show it doesn't work.

Q10. (Mini Project - All Concepts)

Create a **Simple Number Analyzer** program using functions:

Write the following functions:

1. `get_number()` → takes input from user and returns the number.

2. `is_even(num)` → returns True if number is even, else False.
3. `power(base, exp=2)` → returns base raised to exponent (use default argument).
4. `show_result(num)` → prints whether the number is even/odd and its square.

Use a while loop to repeatedly ask the user for numbers until they enter 0.
Use recursion **only** if needed for one small part.
